AI Application Integration Requirement Analysis

Foreword

- This course describes the development status and challenges of AI, and the requirement analysis process during AI application development.

Objectives

- Upon completion of this course, you will be able to:
  - Understand the development and challenges of AI.
  - Understand the AI application requirement analysis process.

Contents

1. **AI Technology Development Status and Technical Challenges**

   1.1 Computer Vision

   1.2 Natural Language Processing

   1.3 Voice Processing

2. AI Application Requirement Analysis

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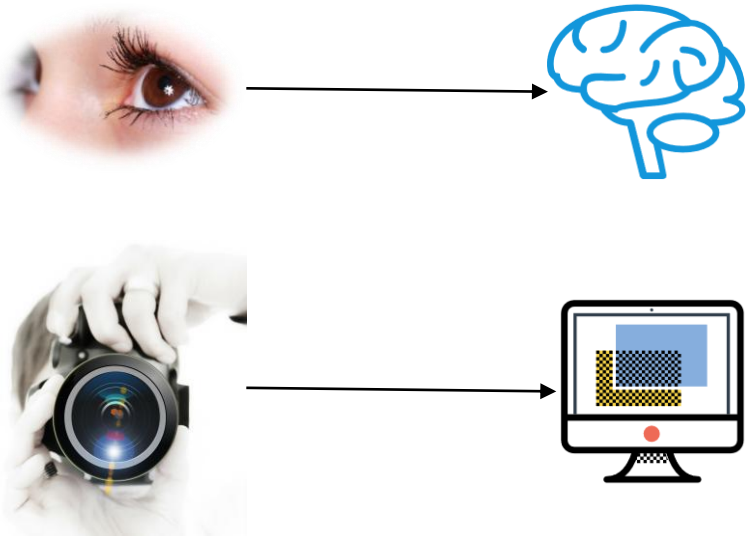
What Is Computer Vision?

- Computer vision leverages artificial intelligence (AI) to allow computers to obtain meaningful data from visual inputs such as photos and videos. CV enables computers to sense the environment by installing cameras and algorithms as eyes and brains.



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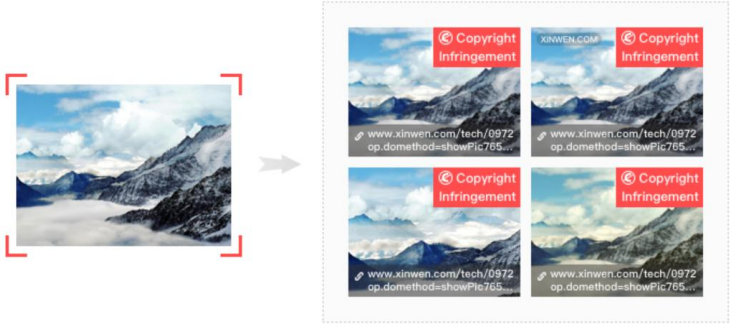
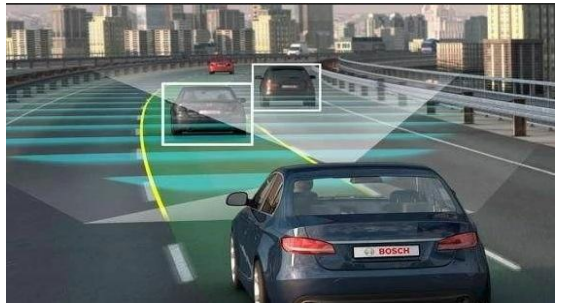
Problems Solved by Computer Vision

- People, object, and scene recognition

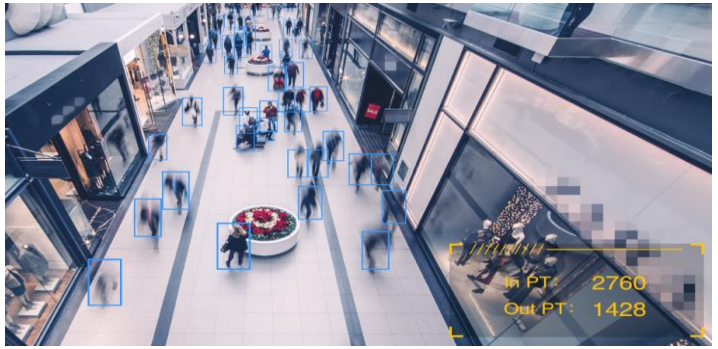
- Search by images



- Detect and avoid obstacles



- Bird's eye view



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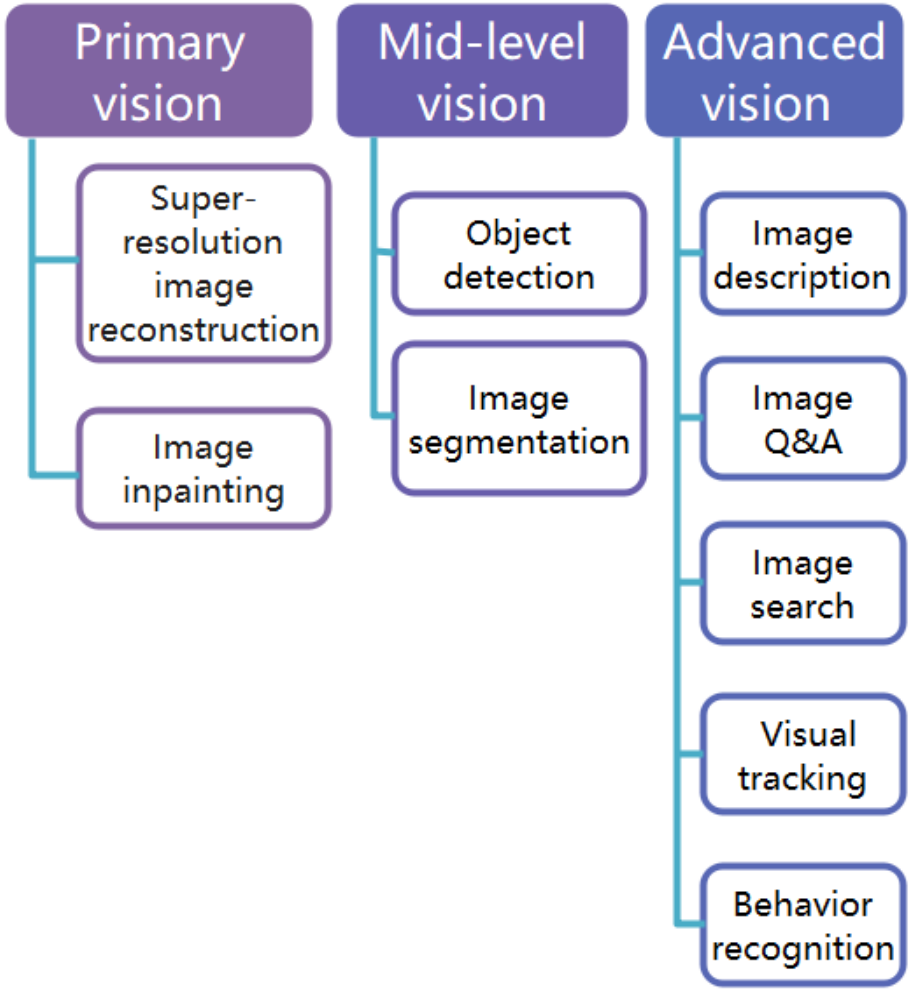
Computer Vision Task Classification



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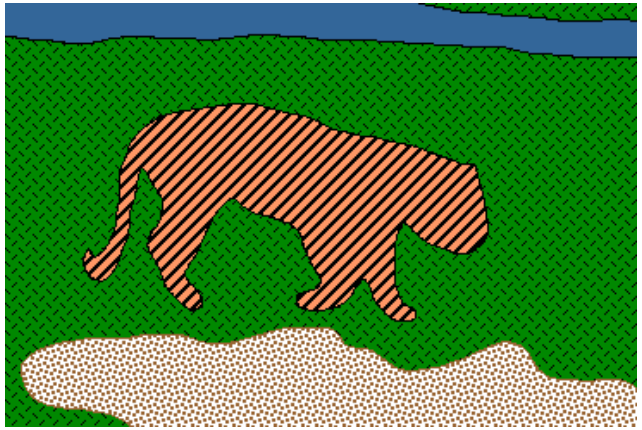
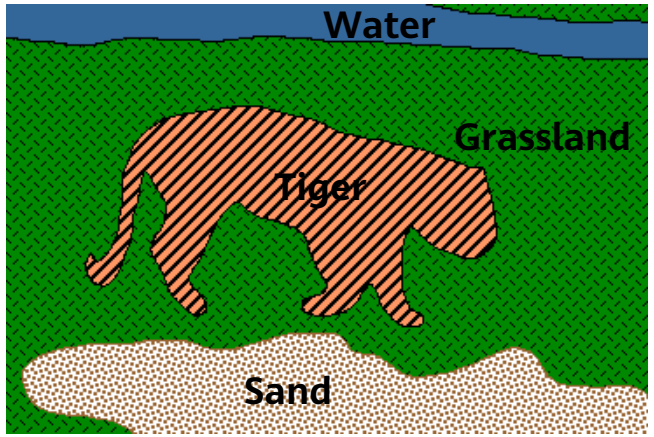
Three Levels of Computer Vision

Pixel

Edge, direction, and texture



Object and scene



Foundation → Middle level → Advanced

Image processing

Image analysis

Image understanding

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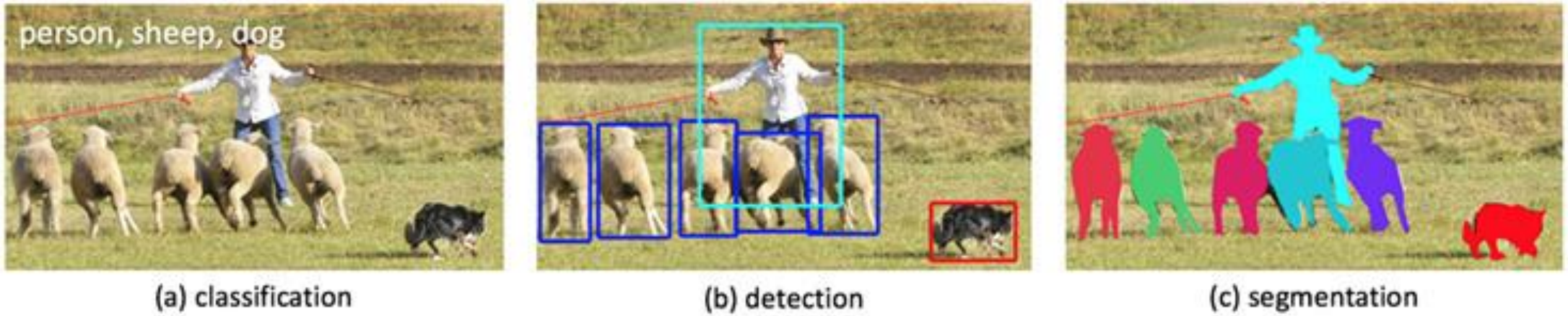
Mid-Level Vision Tasks

- Mid-level computer vision tasks:
  - Classification: determines the type of objects contained in an image or a video.
  - Detection: locates and identifies the objects.
  - Segmentation: composes instance and semantic segmentation, and determines which object or scene each pixel belongs to.



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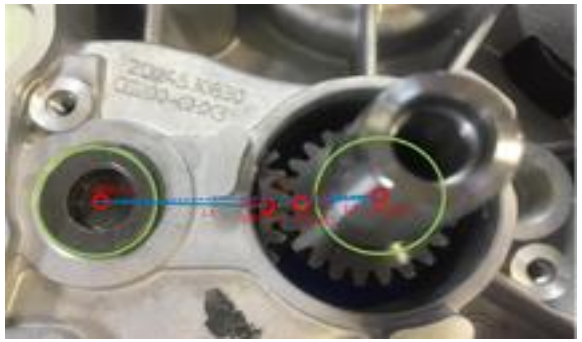
Application Scenarios of Image Classification

Content detection

Security check



Quality test

Parts classification

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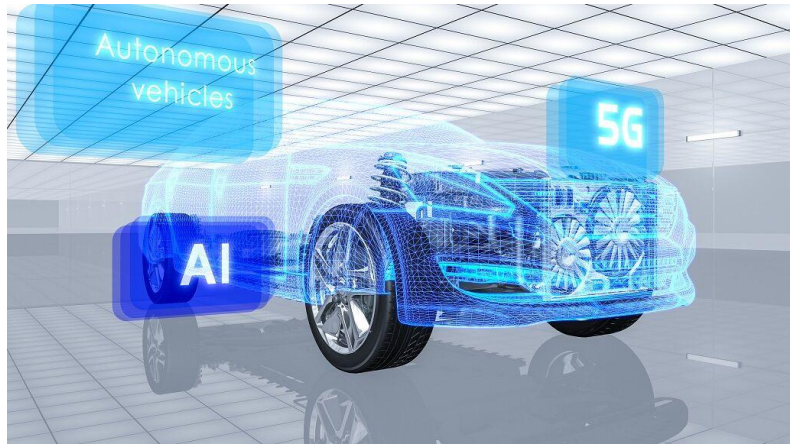
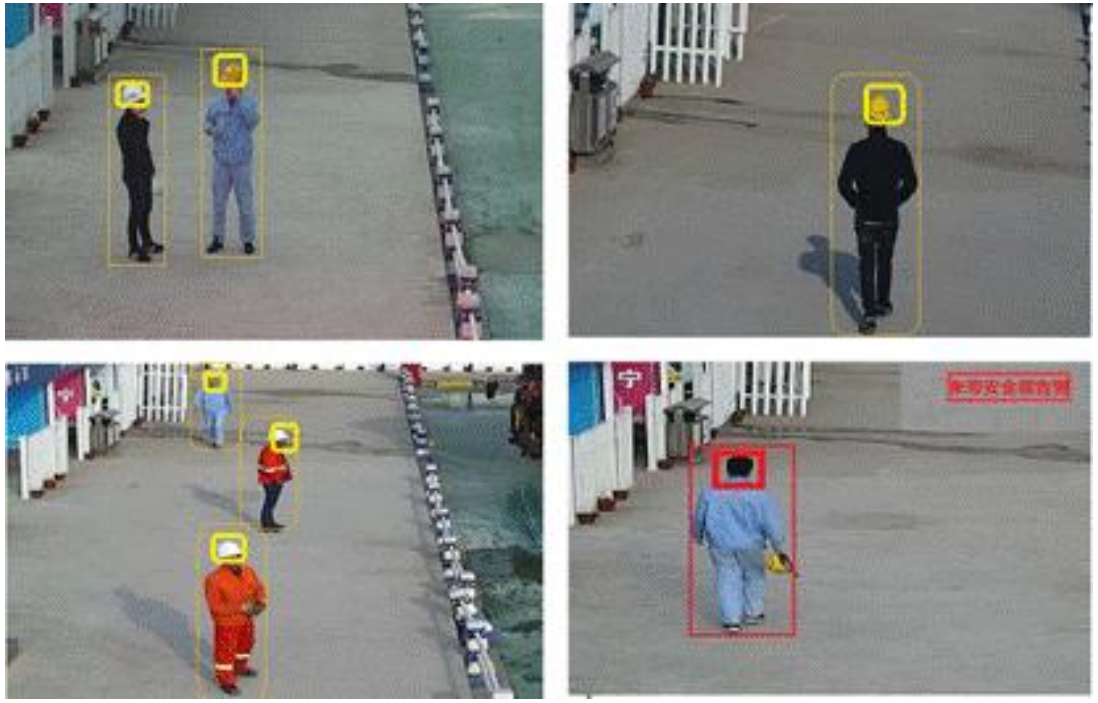
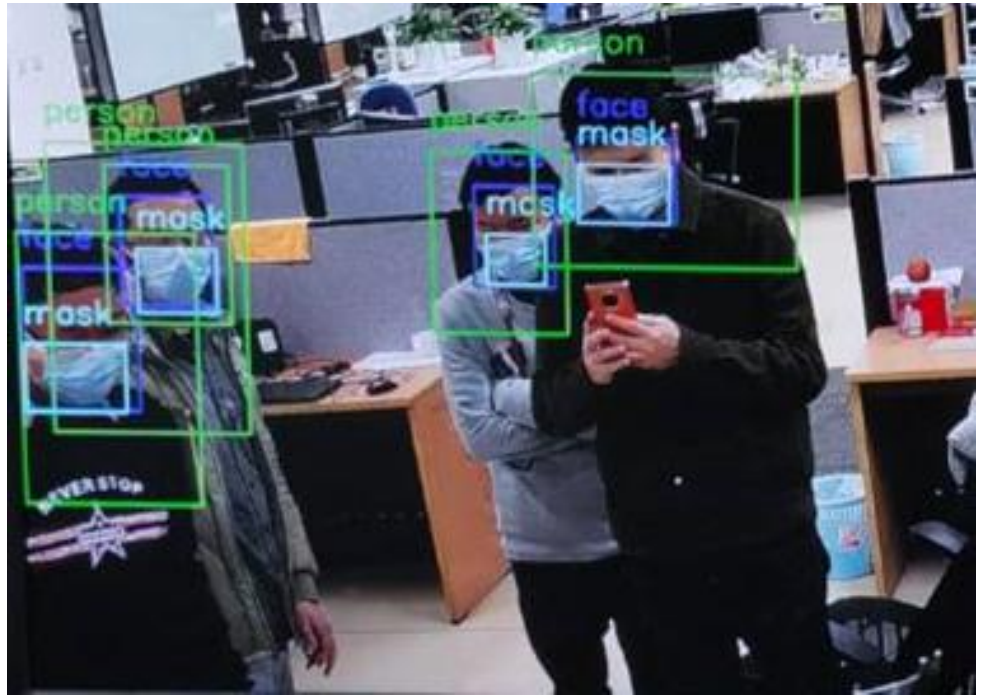
Application Scenarios of Object Detection (1)

- The object detection technology is used in many scenarios:
  - In autonomous driving, AI automatically detects and helps vehicles to avoid obstacles such as vehicles and pedestrians on roads.
  - In the medical field, AI detects and identifies abnormal cells in case tissues to assist doctors in diagnosis.
  - In the coal mining field, AI checks whether the equipment is abnormal and whether the personnel have dangerous behaviors, and assists in safety helmet detection, people intrusion detection, hydraulic support detection, and belt object detection.



Application Scenarios of Object Detection (2)





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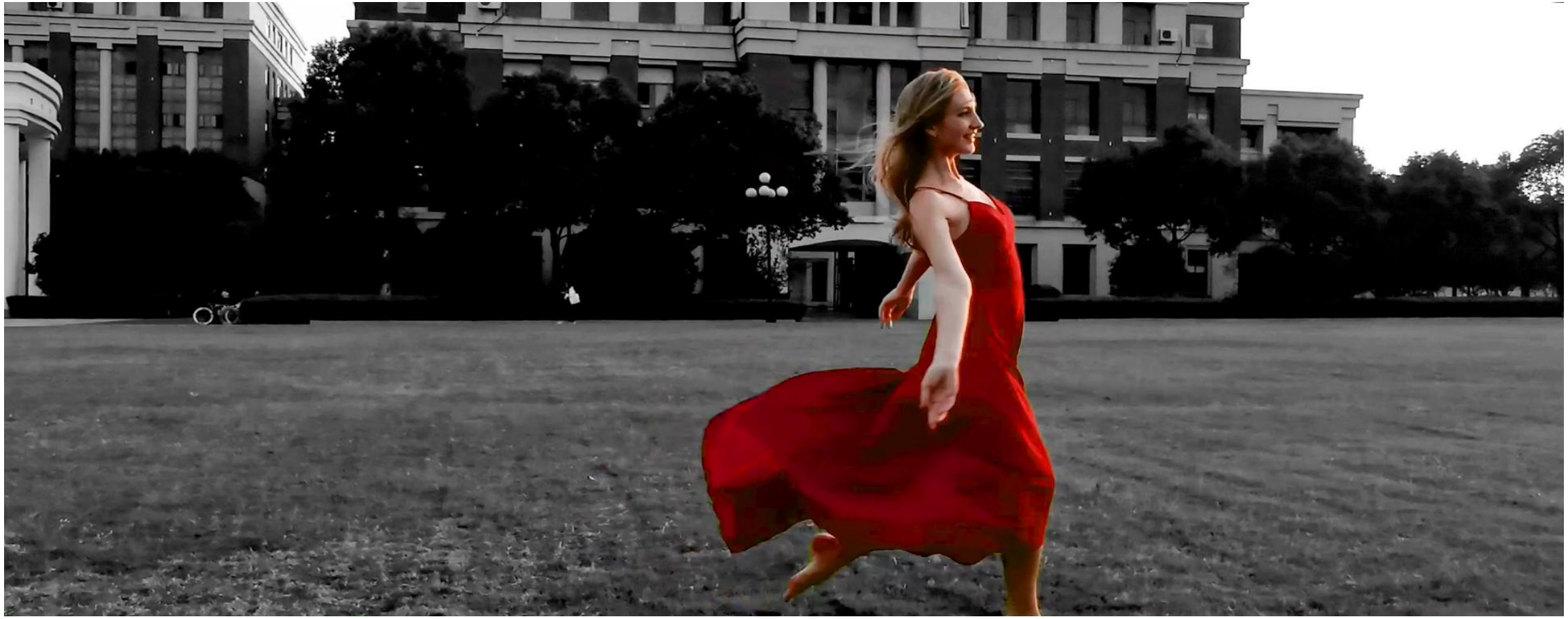
Application Scenario of Image Segmentation



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Computer Vision Challenges

Challenge	Example
Illumination change	Overexposure and underexposure
Scale variation	Two objects of the same physical size would appear as different size in an image. For example, the same vehicle may take up more than 80% or less than 10% of an image.
Blocked objects	Some pedestrians, vehicles, and bicycles are blocked.
Posture variation	People have various postures, such as standing, bending over, squatting, and lying.
Motion blur	When an object moves too fast, it is blurry in a video image.
In-plane rotation	A book is placed upside down or sideways.
Hyperplane rotation	Head nod and shake detection
Background interference	A fish is swimming over cobblestones.
Low resolution	After a 10 x 10 vehicle in an image with a resolution of 1024 x 1024 is resized to 256 x 256, the resolution of the vehicle is only 2 x 2.

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What Is NLP? (1)

- Natural Language Processing (NLP) is a technology that uses computers to process and analyze text and spoken words in much the same way human beings can.

Natural language processing = Natural language + Processing

Human language: It is a symbol system consisting of words and grammar, with speech as the material shell. Text and sound are two attributes of a language.

Machine: A machine can process human languages so that it can understand and generate human languages.

  - NLP is a theory and technology that uses computers to process human languages, enabling computers to understand and generate human languages.

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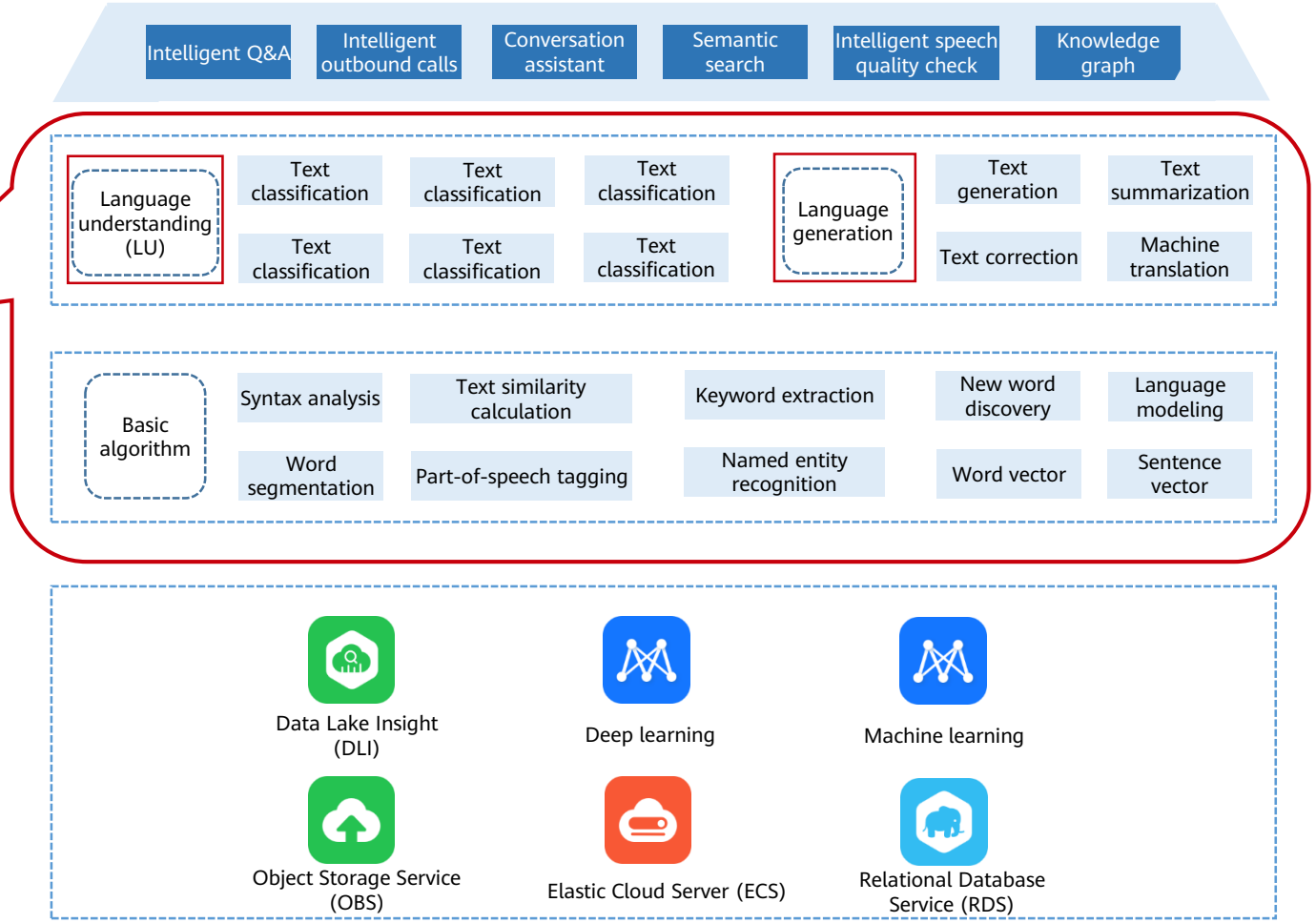
What Is NLP? (2)

Natural language + Processing



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NLP Application Scenarios

Intelligent Q&A	Public sentiment analysis	Text classification	Entity extraction	Summarization
AI poetry writing	Machine translation	Content moderation	Inter-text translation	...

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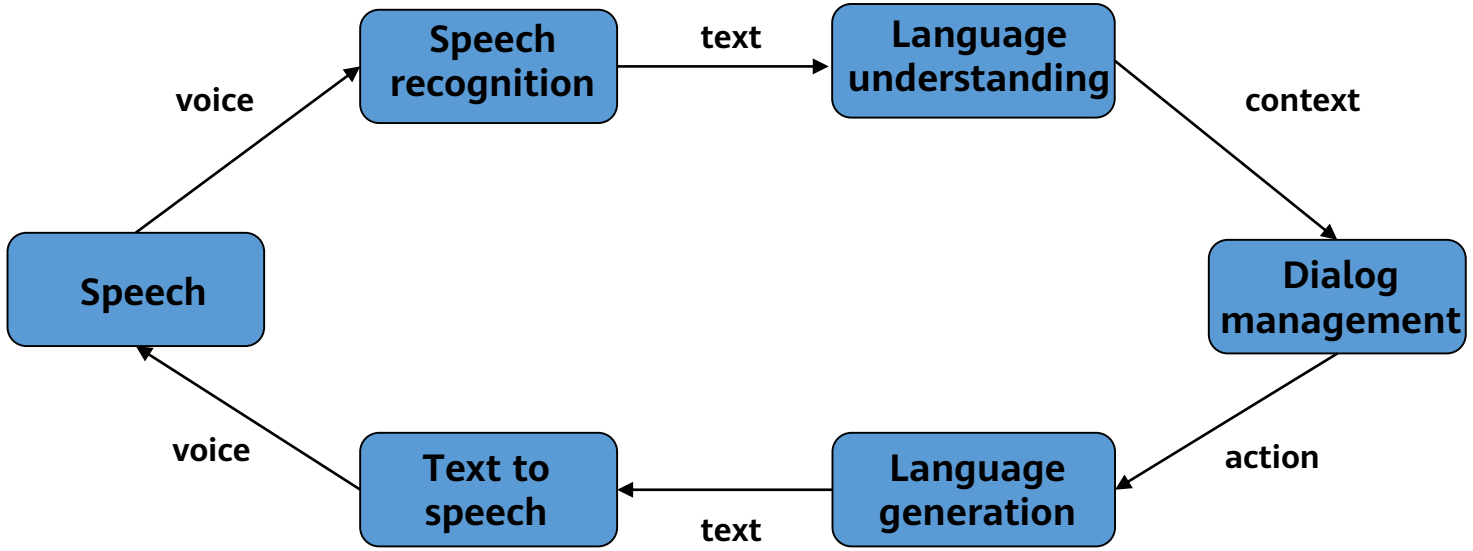
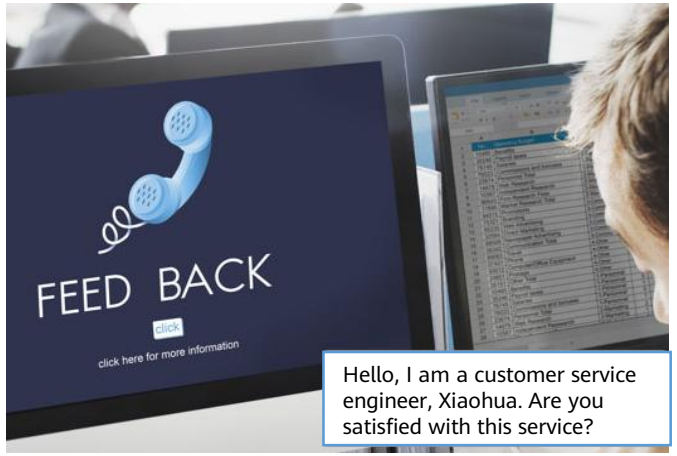
What Is Voice Interaction?



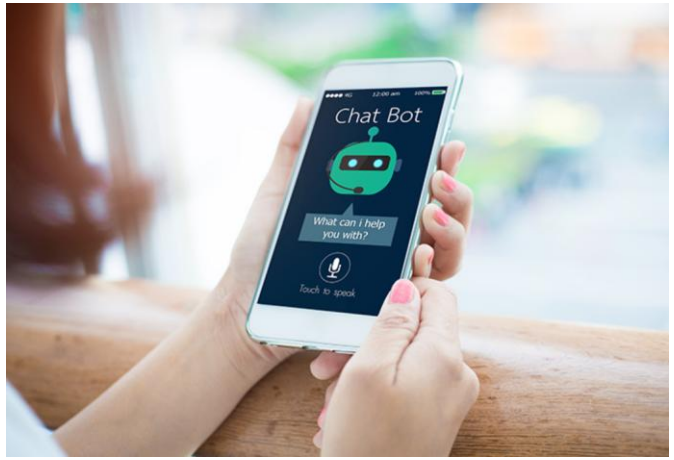
Application Scenarios of Speech Interaction

Live captions

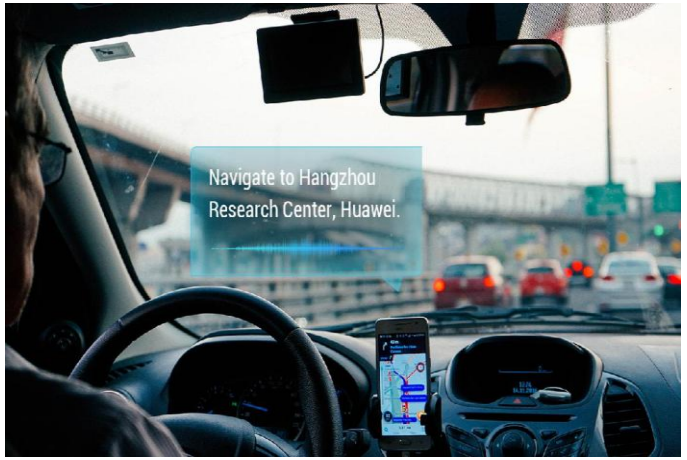
Meeting minutes



Telephone follow-up



Voice search



Voice navigation

Audiobook narration

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Challenges in Speech Interaction

Challenge	Description
Input variation	Different people have different vocal cord conditions, accents, and speaking styles. People may speak in different emotions, at different times, or in different physical conditions.
Background noise	Background noise Transport channel and microphone quality
Model validity	Inadequate training data and impact of dirty data High output dimensions and large search space Difficult to integrate external knowledge such as psychology and linguistics

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Comparison Between Conventional Software Development and AI Application Development

Item	Conventional Software Development	AI Application Development	
Requirement analysis	① Clarify problems and user requirements. ② Analyze the feasibility.	① Clarify problems and user requirements. ② Analyze the feasibility and decide whether to introduce AI and what AI to introduce.	
Product design	Design product prototype.	Design service solution prototype.	
Architectural design	① Generate and break down product requirements. ② Formulate the technical solution. ③ Design the system architecture. ④ Design the modules.	① Generate and break down product requirements (including evaluating AI requirements). ② Formulate the AI technical solution and system technical solution. ③ Design the system architecture. ④ Design the modules.	
Development	① Write and review code. ② Test the units. ③ Review and publish a version.	Algorithm, model, and application development: ① Prepare data. ② Select or develop an AI algorithm. ③ Train and tune an AI model. ④ Repeat the preceding steps to obtain available models. ⑤ Orchestrate models to generate AI applications.	System development: ① Write and review code. ② Test the units. ③ Review and publish a version.
Test	Integration testing, system testing, and acceptance testing	Integration testing, system testing, and acceptance testing (on systems and models)	
Release	① Software release ② Documentation Release	① AI application release ② Documentation release	
O&M	① Software maintenance ② Documentation maintenance	① AI application maintenance (especially data and model maintenance) ② Documentation maintenance	

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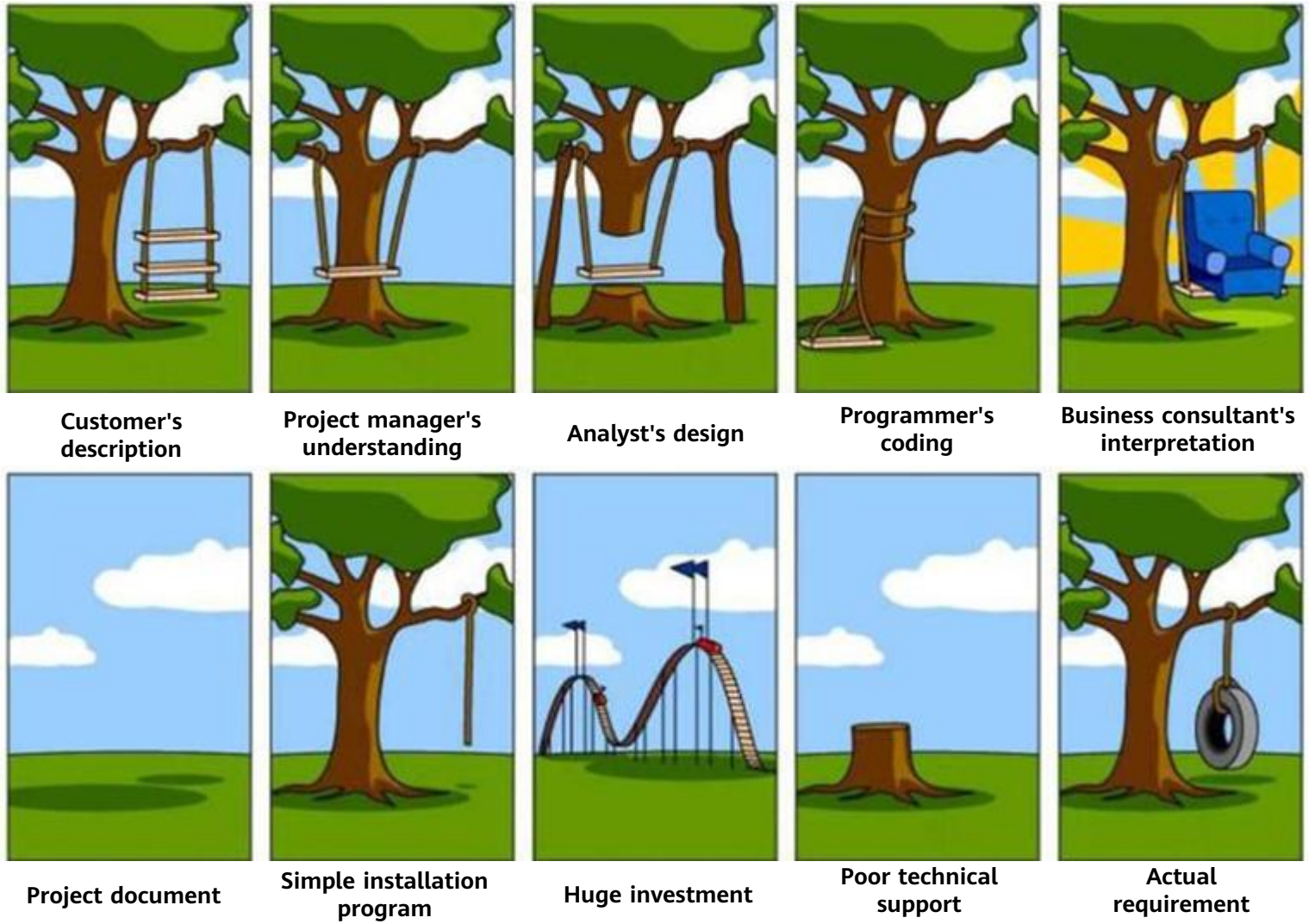
Why Do We Need Requirement Analysis?



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Requirement Analysis

Background	What problems are encountered in what scenarios? Why does the customer want to use AI to solve this problem?
Benefits	Why does the customer want to solve the problems? What benefits can be brought by solving the problems?
Description	How does the customer expect the problems to be solved? What are the key service metrics?
Problem abstraction	Convert the service problems into AI technical problems by modeling and abstraction.
Feasibility analysis	Is there available data? Is there any AI algorithm for the problems? Can the precision/performance meet expectation? Is the cost acceptable?

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Practice: Intelligent Body Temperature Monitoring System

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Background

- Prevention and control of infectious diseases require body temperature monitoring. In places with high foot traffic, such as airports and railway stations, conventional temperature measuring is not efficient. So an intelligent and contactless temperature measuring and monitoring solution is required for such places.
- The infrared imaging technology can measure the temperature of each object in an image captured by a camera, but cannot directly tell the temperatures of human bodies from those of other objects. Human bodies can be recognized in an image by human eyes. However, it is impossible to monitor a large number of people 24/7 only by manual labor. So AI is needed to do the job.

Requirement Value

- In densely populated places such as campuses, offices, primary and secondary schools, and universities, AI is used to implement no-contact temperature measurement in multiple scenarios. It greatly reduces the workload and quickly detects people with abnormal body temperature, facilitating the prevention and control of infectious diseases.

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Requirement Description

- Thermal imaging cameras and common HD cameras need to be deployed in public places to quickly locate people with abnormal body temperature in the certain area and trigger alarms.
- The location of all people must be identified and the body temperature must be measured within 1 second. The temperature measurement error must be no more than 0.3°C.

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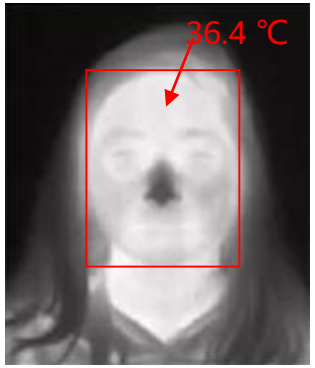
Problem Abstraction

- Convert the service problems into AI technical problems by modeling and abstraction.

1. Find a person from a visible light image.

2. Select the area of exposed skin from the visible light image.

3. Compare the visible light image with the thermal image, and find the corresponding area in the thermal image.



4. Calculate the temperature of the area using thermal imaging software.

How does the software automatically monitor body temperature?

Map human behavior to specific AI technical fields.

Computer vision Pedestrian detection	Computer vision Face detection	Computer vision Image alignment	Non-AI technology Thermal imaging for temperature measurement

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Feasibility Analysis

	Data	Algorithm	Precision/Performance Metric	Cost
Face detection	No dataset needed for face detection model training	There are multiple suppliers that can provide commercial face detection algorithm.	The face detection algorithm is as accurate as human eyes.	This algorithm requires GPUs to run. • The face detection cloud service is charged based on calling times. • Deploying face detection at edge devices requires hardware purchase fees.
Image alignment	N/A	The open-source OpenCV library has the algorithm.	The algorithm is mature and can accurately align a visible light image with a thermal image within 1 second.	Traditional PCs can run this algorithm.

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Section Summary

- This course describes three main technical directions of AI: computer vision, natural language processing, and speech processing. It focuses on the definition, development status, and challenges of each direction.
- This course also describes the requirement analysis process of AI application development. Requirement analysis is an important step in AI application development and includes background analysis, benefits analysis, requirement description, requirement abstraction, and feasibility analysis.

Q&A

1. Which of the following are computer vision tasks? ()

   A. Image classification

   B. Object recognition

   C. Sentiment analysis

   D. Image segmentation

   E. Super-resolution image reconstruction

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Q&A

2. Which of the following steps are included in the requirement analysis of AI application development? ()

A. Background

B. Requirement benefits

C. Requirement description

D. Requirement abstraction

E. Requirement source

F. Feasibility analysis

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Thank You.

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